

Introduction to Computer Simulation

Home Work 2

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For question number one the following code was used:

```
import numpy as np

def get_states(N):
    #Creates a string list with all the possible combinations in binary
    a = [list(bin(i).split('b')[1].zfill(N)) for i in range(2**N)]
    indx = np.zeros((len(a),len(a[0])))

    #Converts the list into an np array with integers
    for i in range(0,len(a)):
        for j in range(0, len(a[0])):
            indx[i][j] = int(a[i][j])

    #Transforms the binary array such that 0 -> 1 & 1 -> -1
    return indx*(-2)+1

def get_energy_all_states(N,J):
    #set necessary arrays
    state = get_states(N)
    Energy = np.zeros(len(state))

    for j in range(0,len(state)):
        Energy_j = 0

        #Assigns the energy of state j into the jth element of the Energy array
        for i in range(0,len(state[0])-1):
            Energy_j += state[j][i]*state[j][i+1]

        Energy[j] = Energy_j
```

```

    #Multiplies all the energies by -J
    #Also returns all the unique energies for all possible states
    return -J*Energy , np.unique(-J*Energy)

def get_Partition_Function(N, J, beta_max, step = 0.01):
    #Gets the Energies of all possible states
    E , E_unique = get_energy_all_states(N, J)

    #All possible beta values in the step resolution (excluding 0)
    beta = np.arange(step, beta_max+step, step)

    #Creates an array with the sum of  $e^{-\beta E}$  for each value of beta
    Z = np.sum(np.exp(-np.outer(beta, E)), axis=1)

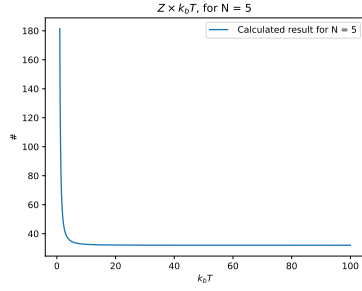
    #Returns all the relevant variables
    return Z , beta, E, E_unique

def get_Probability(E, Z, beta, E_indx):
    #Calculates the degeneracy associated to the energy level E_indx
    degeneracy = np.count_nonzero(E_indx == E)

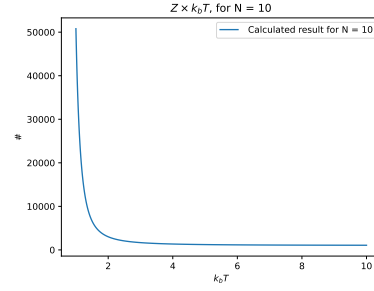
    #Returns the probability distribution as a function of beta
    return degeneracy*np.exp(-beta*E)/Z

```

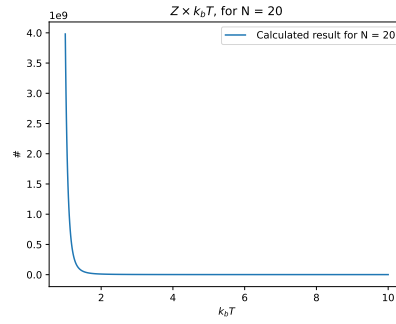
With this we can obtain the Partition function as a function of temperature, i.e. $k_b T = 1/\beta$, with this we get the following graphs:



(a) $Z \times k_b T$ for $N = 5$



(b) $Z \times k_b T$ for $N = 10$



(c) $Z \times k_b T$ for $N = 20$

Figure 1: Partition function for $N = 5, 10, 20$.

We can see that the Partition functions decrease as the Temperature increases, showing that in low temperatures the terms that minimize the energy dominate. For $\beta = 1$, the following values were found for Z

$$Z(N = 5, \beta = 1) = 181.43$$

$$Z(N = 10, \beta = 1) = 50792.26$$

$$Z(N = 20, \beta = 1) = 3980922408.72$$

2

2.1

From the definition of the free energy per spin and the relation that:

$$\beta f_{\infty} = -\ln(2) - \ln(\cosh(\beta)) \quad (1)$$

it follows that:

$$\beta f_N - \beta f_{\infty} = \ln(Z^{1/N}) + \ln(2) + \ln(\cosh(\beta))$$

Using the properties of logarithms we arrive at:

$$\Delta(\beta f) = \ln(2\cosh(\beta)Z^{1/N}) \quad (2)$$

The solution of the 1-D Ising model with free boundary conditions is:

$$Z(\beta, N) = 2^N \cosh^{N-1}(\beta) \quad (3)$$

Combining this with equation (2), we get:

$$\Delta(\beta f) = \ln(4\cosh^{\frac{2N-1}{N}}(\beta)) \quad (4)$$

Similarly for periodic boundary conditions we use the fact that the Partition function is:

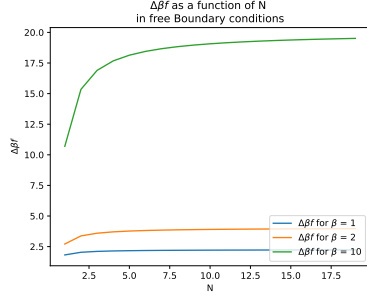
$$Z(\beta, N) = (2\cosh(\beta))^N + (2\sinh(\beta))^N \quad (5)$$

Plugging this into (2), we get:

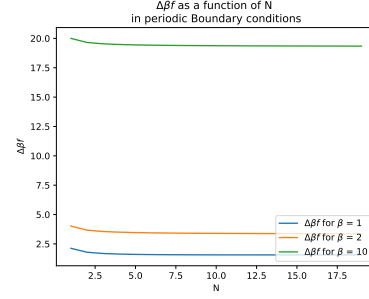
$$\Delta(\beta f) = \ln(4\cosh(\beta)) + \frac{1}{N}\ln(\cosh^N(\beta) + \sinh^N(\beta)) \quad (6)$$

2.2

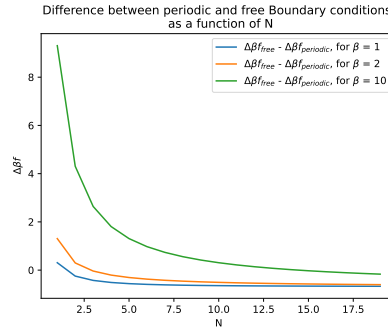
Plotting the functions:



(a) $\Delta(\beta f_{free})$ for multiple β



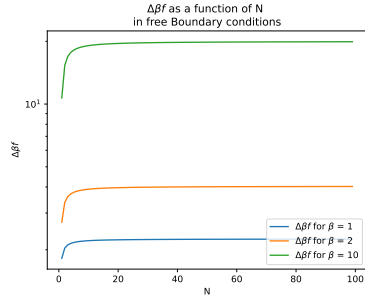
(b) $\Delta(\beta f_{periodic})$ for multiple β



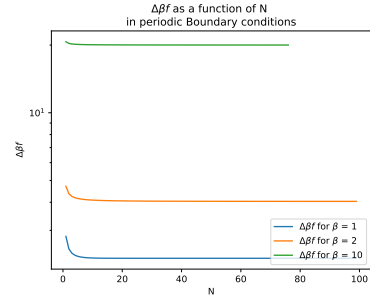
(c) $\Delta(\beta f_{free}) - \Delta(\beta f_{periodic})$ for multiple β

Figure 2: Plots of the free energy correction for different boundary conditions

The Corrections quickly approach a constant value, as N increases, we can also notice that this constant value is proportionally related to β , the difference for the two boundary conditions decreases rapidly as N increases.



(a) $\Delta(\beta f_{free})$ for multiple β



(b) $\Delta(\beta f_{periodic})$ for multiple β